Answer **all** questions.

1. The Krebs cycle is a set of enzyme-controlled reactions that take place in aerobic organisms. **Image 1.1A** shows a simplified diagram of the Krebs cycle and **Image 1.1B** shows the structural formulae of two key intermediates.

Image 1.1A

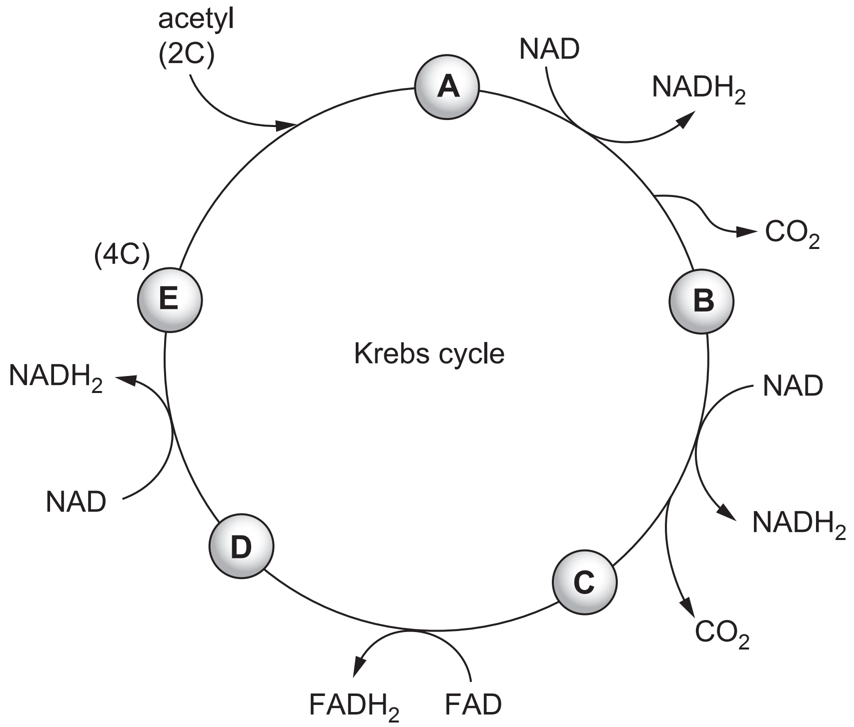

Image 1.1B

α ketoglutaric acid

citric acid

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(a) (i) Complete **Table 1.2** by recording the number of each type of atom present in the two named compounds shown in **Image 1.1B**. [2]

Table 1.2

Type of atom	Number of atoms present	
	α ketoglutaric acid	citric acid
Carbon	....	....
Hydrogen	....	....
Oxygen	....	....

(ii) Using the information provided in **Images 1.1A** and **1.1B**, identify which of the intermediates labelled **A–E** on the drawing of the Krebs cycle is α ketoglutaric acid and which is citric acid. [1]

α ketoglutaric acid ....

citric acid ....

(iii) Use the information in **Image 1.1A** to state what happens to each of the following atoms in the conversion of **A** to **B**. [2]

Carbon ....

....

Hydrogen ....

....

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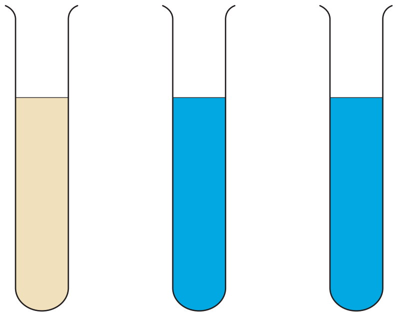
(b) Malonic acid is a dicarboxylic acid with the chemical formula $C_3H_4O_4$. **Image 1.3** shows a summary of an experiment to test the hypothesis that malonic acid is a respiratory poison in yeast.

Methylene blue is used to monitor **dehydrogenase** activity because it acts as a hydrogen acceptor and turns from blue to colourless when it is reduced.

Image 1.3



Tube	1	2	3
Boiled and cooled yeast suspension (cm^3)	0	0	10
Active yeast suspension (cm^3)	10	10	0
Malonic acid solution (cm^3)	0	5	0
Water (cm^3)	5	0	5
Methylene blue solution (cm^3)	1	1	1
Colour after 30 mins	cream	blue	blue

(i) Tube **3** in **Image 1.3** acts as a control. Describe the purpose of tube **3** in this experiment. [1]

....

....

(ii) Describe the evidence from **Image 1.3** which supports the hypothesis that malonic acid is a respiratory poison. [2]

....

....

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....

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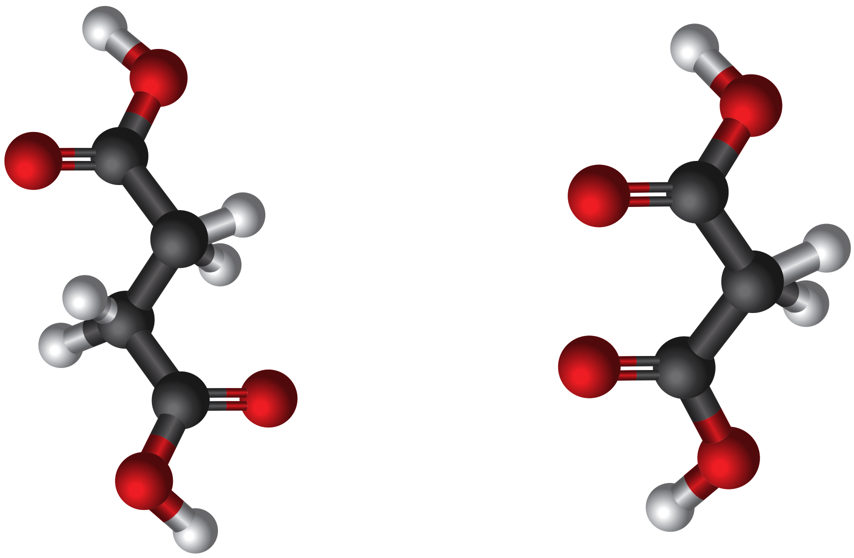
(iii) **Image 1.4** shows ball and stick models of another Krebs cycle intermediate, succinic acid, and a molecule that is believed to act as a respiratory poison, malonic acid.

Image 1.4



succinic acid malonic acid

Using information from **Image 1.4** and your knowledge of enzymes suggest how malonic acid could act as a respiratory poison. Explain your answer. [2]

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